# Full Text, Language, and Embeddings

## Full Text, Language, and Embeddings I

Beyond bibliographic metadata, the pipeline mines the textual content of each record. This stage bridges the gap between a bibliographic inventory and a semantic understanding of the literature.

## Full-Text Availability I

An open-access resolver maps each record to a downloadable PDF where one exists (a known `pdf_url`, or an Unpaywall lookup by DOI), and an opt-in, network-gated downloader fetches it to a deterministic path. Full-text availability is summarized without requiring any download, so the offline default still reports coverage. For this run:

# Full-Text Availability I

- **Abstract coverage**: 100.0% of records (80 of 80 retrieved records) carry an abstract; 0 records lack one.
- **Open-access status**: 42.5% of records are open access (34 records); the remainder are closed or unknown.
- **PDF availability**: 42.5% of records (34) have a direct PDF link; 34 have a publisher PDF, and 46 have no full-text source available.

## Full-Text Availability I

The identifier coverage for this corpus is: 80 DOIs, 12 OpenAlex IDs, and 0 arXiv IDs. DOI coverage dominates and supports cross-engine de-duplication.

## Language and Entity Extraction I

Titles, abstracts, and (when present) full text are tokenized and reduced to keyphrases and named entities by offline, dependency-light extractors — no mandatory LLM. Term-frequency statistics drive a TF-IDF representation over a 137-feature vocabulary. The most frequent terms in the corpus are: atmospheric, uncertainty, spectra, temperature, abundance, analysis, assumptions, opacity, retrievals, sources, stellar, coverage, evaluated, observed, across, composition, models, provide, competing, retrieval. These terms reflect the configured domain and the records retained by the retrieval and filtering policy; fixture-derived terms are not evidence about the live field.

## Embeddings I

Every title, abstract, and full text is embedded into a shared vector space by a deterministic, offline method — TF-IDF followed by truncated SVD, i.e. latent semantic analysis (Deerwester et al. 1990). The embedding dimensionality is 50 components (configurable via `project_config.embeddings.n_components`), and the TF-IDF vocabulary is capped at 137 features (configurable via `project_config.embeddings.max_features`). The embedding is byte-stable across runs: the same input text always yields identical vectors, so the derived similarity matrix, nearest-neighbour lists, clusters, and two-dimensional projection are all reproducible.

An optional transformer backend can be enabled by setting `projec t_config.embeddings.method: transformer` (requires the `embeddings` extra), which upgrades the embedding fidelity without changing the interface or downstream analysis.

The embeddings support semantic retrieval over the corpus and feed two visualizations: a PCA two-dimensional projection ((Figure pca embeddings)) that maps the topical geography of the literature, and a hierarchical clustering dendrogram ((Figure dendrogram)) that reveals the similarity structure of the document collection.